Analysis of Different Feature Selection Techniques on Different Machine Learning Approaches to Classify Leukemia Subclasses

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***Abstract*—The correct identification of specific cancer subclasses is one of the significant challenges in the treatment of leukemia. Utilizing genomic profiles from microarrays is a useful tool for classifying different cancer subtypes. It is difficult since using vast amounts of gene expression data will demand a lot of processing power. The findings in our study focus on comparing several feature selection techniques in order to identify the genetic expressions that are extremely lethal. The feature selection procedure improves the algorithms' capacity for prediction. The computational cost and data volume can be significantly reduced by choosing the appropriate feature. The correct identification of specific cancer subtypes is one of the major difficulties in treating leukemia. Utilizing genetic profiles from microarrays is a useful tool for classification. To classify Acute Myeloid Leukemia (AML) and Acute Lymphoblastic Leukemia (ALL) on features chosen in three different ways, namely the Filter method, Wrapper method, and Embedded method, six machine learning algorithms, and an Artificial Neural Network was constructed. One process achieved a success rate of 100% and the rest of the procedures procured about 95% accuracy on average.**

***Index Terms*—Microarray, Feature Selection, Filter Method, Wrapper Method, Embedded Method, Machine Learning Algorithm, Artificial Neural Network, ALL, AML**

I. Introduction

Cancer is generally distinguished by abnormal growth of tissues/cells. It can affect almost any body part. Leukemia is a cancer of the body's blood-forming tissues. It is detected when the DNA of a single cell in bone marrow mutates and cannot function properly. Leukemia affects the lymphatic system as well. It is a lethal disease. According to the latest WHO data published in 2020 Leukemia Deaths in Bangladesh have reached 1895 or 0.26% of total deaths [1]. Identification of the proper type of cancer is necessary to determine the required treatment. Thousands of individuals are being victims of its fatality. Two fatal types of Leukemia are Acute Lymphoblastic Leukemia (ALL) and Acute Myeloid Leukemia (AML). If leukemia develops a lymphocyte in bone marrow, then it is called Lymphoblastic Leukemia. If Leukemia evolved within bone marrow cells, red blood cells, and platelets then it is called Myeloid Leukemia. In order to ensure proper treatment, it is very important to detect leukemia subclasses in an efficient manner as treatments vary considerably.

Initially, cancer was detected by morphological means. To acquire more knowledge and efficient classification it is very much important to gain information about underlying intricacy and obscurity. Genetic expression analysis provides profound information and criteria for classifying cancer subtypes. It is also a very reliable procedure as morphological attributes follow a highly deviated range of patterns, while genetic patterns are much easier to detect. Besides, morphological analysis of tumor cells is a tedious and costly procedure. The limits of conventional morphological analysis stimulate the intervention of a profound criterion. To eliminate these barriers gene expression analysis can be a reliable measure. To obtain necessary data DNA microarray can be a significant choice as it can measure thousands of gene expressions simultaneously. Numerous studies have been carried out utilizing gene expression data for methodical cancer classification.

An approach deployed on gene expression profiles extracted from DNA microarray has been proposed by Golub et al [2]. This study acknowledges the underlying correlation among genetic attributes and indicates that leukemia can be classified solely based on genetic profile. Here, in our study, our working pipeline is highly stimulated to analyze effective feature selection in order to conglomerate genes with the most vital information. In this study, we have tried to experiment with three types of feature selection techniques. A variety of machine learning algorithms including an artificial neural network algorithm and ensemble learning was performed to determine the compliance of all the methods in a variety range of accuracy metrics.

In short, the significant contributions of this work are as follows -

- Investigating three different types of feature selection methods.
- Comparison of selected features with the raw train data.
- Implementation of a various learning algorithms to compare the efficiency of multiple techniques.

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II. Related Literature

There have been many studies on gene expression throughout the years thanks to recent developments in artificial intelligence. In these experiments, all share the same objective of improving prediction accuracy, cutting down execution time, and finding patterns as quickly as possible. Authors in [3] suggested tree-based methods for classification. The cancer data was classified using the C4.5 decision tree and bagged and boosted decision trees. Their result depicts that bagging performs better than the other two methods. To improve performance Null space-based feature selection was introduced in [4]. Redundant gene expressions were eliminated by null space. In [5] the authors suggested Random Forest algorithm for the selection of gene patterns. The main objective of their model was to find significant genes with improved accuracy. SVM with optimized parameters for correlated genetic feature extraction was introduced in [6] Artificial Neural Network (ANN) was used for the effective classification of cancer [7]. The classification result of ANN was better compared to other techniques. A deep neural network was used for classification in [8]. Good accuracy was achieved by combining DNN and logistic regression in an ensemble technique. It's evident that an ample amount of research has been performed on gene expression analysis. To increase accuracy and reduce time, a variety of experiments were performed over time.

III. Materials and Methods

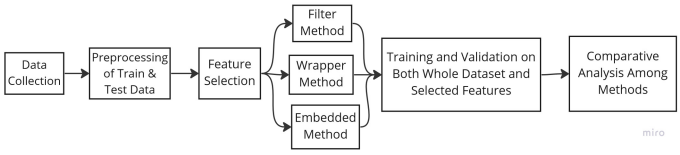


Fig. 1. Working pipeline.

A. The Dataset

The gene expression dataset was first introduced in the study of [2]. The dataset comes from a proof-of-concept study of the DNA microarray experimental method. It is used to extract the genetic expression of cancer cells. The study shows that gene expressions can be very useful for the categorization of different types of cancer cells. These data are highly useful for differentiating Acute Myeloid Leukemia (AML) and Acute Lymphoblastic Leukemia (ALL) based on their genetic characteristics. The Dataset is publicly available at [10] and it is used for this particular study. The data set is consisting of three files:

actual.csv: This file contains a specific patient number and what type of Leukemia a patient is suffering from. Among the total of 72 patients, 47 are suffering from Acute Lymphoblastic Leukemia (ALL) and 25 are suffering from Acute Myeloid Leukemia (AML).

data_set_ALL_AML_train.csv: Contains 38 samples of bone marrow data each of which contains 7129 genetic features.

data_set_ALL_AML_independent.csv: Contains 34 peripheral blood samples each of which contains 7129 genetic features.

The training and test dataset contains information about gene expression, gene accession number, and Raw numeric values which quantifies the amount of influence of a particular gene in a sample. In preprocessing the dataset was transposed and 'call' columns containing 'A' only were dropped, followed by the dropping of all the 'call' columns. After this 5327 gene expressions were retrieved.

B. Feature Selection

1) Filter Method: Here various statistical tests are performed on features [21]. The features are selected based on their score for their correlation with the output variable. It is a machine learning algorithm-independent method.



Fig. 2. Filter methods.

For our study, ANOVA (Analysis of variance) [22] was implemented as a filtering technique to select features. This particular technique inherently uses f distribution for feature selection. It is highly useful for numerical input and categorical outcome data. The formula to acquire f_score using ANOVA is -

TABLE I
Summary ANOVA.

Source	Sum of Squares	Degree of Freedom	Variance Estimate (Mean Square)	F ratio
Between	SS_a	K-1	$MS_B = \frac{SS_B}{K-1}$	$\frac{MS_B}{MS_W}$
Within	SS_w	N-K	$MS_W = \frac{SS_W}{N-K}$	
Total	$SS_T = SS_B + SS_W$	N-1		

2) Wrapper Method: Here a subset of the feature is chosen at first [23], then the selected subset is fed into the learning algorithm. Their performance is analyzed and again new subset is generated until the best-performing subset is acquired. In our study, the forward feature selection technique is used with the Naive Bayes classifier as the inherent learning algorithm and selects 1000 features.

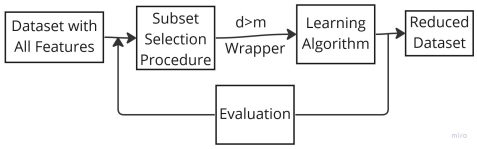


Fig. 3. Wrapper methods.

3) Embedded Method: It combines the benefits of both the filter method and the wrapper method [24]. It performs feature selection and model training in parallel.

In our study, the random forest algorithm was used as an inherent tree-based method. In this procedure, a score is assigned to each feature and the highest-scoring subset is chosen.

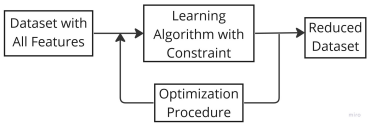


Fig. 4. Embeded methods.

C. Algorithms Used

In this study, the problem is identified as a supervised machine learning classification problem so the following algorithms were used throughout the process.

1) Logistic Regression: Logistic regression [11] is an appropriate algorithm to perform when the dependent variable outcome is binary. This machine learning approach is helpful for classification issues. Data are described and the link between two things are explained using logistic regression.

2) Support Vector Machine (SVM): Support Vector Machine (SVM) [12], [13] is a Supervised Machine Learning algorithm. It is used to solve both classification and regression problems. The main objective of an SVM algorithm is to create a hyperplane to classify dependent variables according to their respective patterns [14].

3) K-Nearest Neighbor: The k-nearest neighbors [15] (KNN) algorithm is a data classification method for estimating the likelihood that a data point will become a member of one group or another based on what group the data points nearest to it belong to.

4) Random Forest Algorithm: Random Forest Algorithm is based on multiple decision trees [16]. Decision Trees solve problems using information gain and entropy of an attribute. The forest algorithm provides output based on the prediction from Decision Trees.

5) Naïve Bayes Algorithm: Naïve Bayes [17] is a probabilistic algorithm. A Naïve Bayes instance predicts an incident based on conditional probability. Features that are not related to one another are incorporated into this algorithm. This classification consists of complimentary conditional assemblies that resemble networks.

6) Adaptive Boosting: Adaptive boosting or ADA boost [18] is an ensemble learning algorithm. Combining with a weak model can result in a powerful one. The principle behind boosting algorithms is first we built a model on the training dataset, then a second model is built to rectify the errors present in the first model. This procedure is continued until and unless the errors are minimized, and the dataset is predicted correctly.

7) Ensemble Learning: A general meta-approach to machine learning seeks better predictive performance by combining the predictions from multiple models. In this study with classical Machine Learning approaches adaptive boosting is used for powerful prediction.

8) Artificial Neural Network (ANN): ANN [19], [20] is inspired by the biological network of the human brain. It is a deep learning method. In artificial neural network architecture lies an input layer, output layer, and a stack of hidden layers. Each layer consists of multiple neurons. The output layer contains the same number of neurons as the class variables. Due to the presence of neurons sometimes ANN is referred to as Multi-Layer Perceptron.

Some preprocessing was performed before applying Machine Learning algorithms. The test and train datasets were transposed so that all the genetic expressions can be used as features. The call columns were matched with respective patient numbers and all the 'call' columns containing only the value 'A' was dropped from the dataset. After this preprocessing both train and test sets had 5327 features and 38, 34 samples respectively. Then these two sets were further processed and split into train and test sets with random sample allocation on every execution. Both test and train sets were normalized to standard scaling for further analysis. ALL AML values of the actual.csv file were label encoded to numeric values 0 and 1 respectively.

IV. Experimental Results and Analysis

The Intel(R) Core(TM) i5-8265U CPU, 8.00GB RAM, and Intel(R) UHD Graphics 620 GPU were used in this experiment. As stated in the methods section several learning algorithms were implemented on the train data set as well as selected features from various feature selection methods. The train test split for this study is 70:30. All the implementations were performed on 10-fold and leave-one-out cross-validated data during model training. Accuracy criterion f1-score, precision, recall, and accuracy were calculated using proper formulae.

Summarized accuracy metrics are depicted through tabular representation for each algorithm.

TABLE II
Logistic Regression.

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.95	0.96	0.95	0.95
ANOVA method	0.86	0.89	0.86	0.85
Forward Feature Selection	0.95	0.96	0.95	0.95
Tree Based method	0.86	0.89	0.86	0.85

Learning Parameters: Regularization = L2, Inverse Regularization Strength (C) = 1, solver = lbfgs, Random state = none, Standardized = False

The performance of features selected by the Forward Feature Selection method is as good as whole train dataset features. Accuracy metrics of ANOVA and random forest are less than no feature selection procedure. Here In logistic regression, the accuracy metrics show an anticipated pattern.

TABLE III
K Nearest Neighbor (KNN).

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.86	0.89	0.86	0.85
ANOVA method	0.91	0.92	0.91	0.90
Forward Feature Selection	0.91	0.92	0.91	0.90
Tree Based method	0.77	0.76	0.77	0.74

Learning Parameters: n_neighbors= 5, weights= uniform, algorithm = auto, leaf size = 30, standardized = False

In the KNN algorithm features selected by ANOVA and Forward Feature Selection perform slightly better than the no feature selection method. The KNN algorithm assumes that similar points share similar labels. Unfortunately, in higher dimensions, highly dispersed points tend to be never close together. So in presence of a vast range of random information, KNN might perform inaptly compared to lesser attributes.

TABLE IV
Support Vector Machine (SVM).

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.82	0.85	0.82	0.78
ANOVA method	0.95	0.96	0.95	0.95
Forward Feature Selection	0.95	0.96	0.95	0.95
Tree Based method	0.91	0.92	0.91	0.90

Learning Parameters: Regularization parameter (C) = 1, kernel = rbf, gamma = scale, decision funtion = ovr, standardization = True

When there are more features than samples, the SVM performs poorly. Performance measurements drop when greater data samples have more noise and outliers present.

TABLE V
RANDOM FOREST.

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.91	0.92	0.91	0.90
ANOVA method	0.86	0.89	0.86	0.85
Forward Feature Selection	0.95	0.96	0.95	0.95
Tree Based method	0.91	0.92	0.91	0.90

Learning Parameters: Standardization = False, n_estimators = 100

In the Random Forest algorithm classification is made based on the information gain calculated from the information entropy of each feature. Here in our work forward feature selection performs better than raw train data. This anomaly can be explained by the trait of the forward feature selection method. This particular method finds out the best possible subset of a fixed length by filtering and testing in a learning algorithm. This method trials every possible subset configuration until it finds the best performing one. So features with larger impact are usually selected by the underlying learning algorithm and tend to have larger information gain. Hence resulting in better accuracy metrics.

TABLE VI
ADAPTIVE BOOSTING.

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.95	0.96	0.95	0.95
ANOVA method	0.95	0.96	0.95	0.95
Forward Feature Selection	0.91	0.91	0.91	0.91
Tree Based method	0.95	0.96	0.95	0.95

Learning Parameters: n_estimators=100, random_state=0, Standardization = True, learnin_rate = 1.0

The accuracy metrics pattern in adaptive boosting does not depict any wearisome anomaly.

TABLE VII
ENSEMBLE METHODS.

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.95	0.96	0.95	0.95
ANOVA method	0.95	0.96	0.95	0.95
Forward Feature Selection	0.91	0.91	0.91	0.91
Tree Based method	0.91	0.91	0.91	0.91

In this method, Logistic Regression was used with Adaptive Boosting.

Learning Parameters: Logistic Regression: Regularization = L2, Inverse Regularization Strength (C) = 1, solver = lbfgs, Random state = none, Standardized = False
Adaptive Boosting: n_estimators=100, random_state=0, Standardization = True, learnin_rate = 1.0

In the ensemble learning method, multiple learning algorithms are used together to obtain better performance. The pattern of accuracy metrics depicts the same.

TABLE VIII
ARTIFICIAL NEURAL NETWORK (ANN).

Methods	Accuracy	Precision	Recall	F1
No Feature Selection	0.95	0.96	0.95	0.96
ANOVA method	1	1	1	1
Forward Feature Selection	0.95	0.96	0.95	0.96
Tree Based method	0.95	0.96	0.95	0.96

Learning Parameters: Input layer: Number of neurons: 500, Activation Function: Sigmoid
Hidden layer 1: Number of neurons: 150, Activation Function: Sigmoid
Hidden layer 2: Number of neurons: 30, Activation Function: Sigmoid
Hidden layer 3: Number of neurons: 10, Activation Function: Sigmoid
Output layer: Number of neurons: 2, Activation Function: Softmax
Loss function: 'sparse_categorical_crossentropy'

In ANN implementation ANOVA depicts a perfect score. ANOVA selects features by comparing variances of features. In our study, In the ANOVA procedure, the features containing less or greater p-value than critical values were eliminated. As a result, the method extracted features from the dense region which fall within critical values only. So, this is a reasonable explanation for perfect accuracy metrics.

After preprocessing 5327 numbers of features existed in the train data set. Among three feature selection techniques, Using ANOVA around 2700+ features are extracted in each execution. In the Forward Feature Selection, it is 1000 features in iteration and in Random Forest it is 220+ features in each iteration. ANOVA and Forward Feature Selection depicts promising performance in most of the learning algorithms. Since the number of features in Random Forest selection is very low the accuracy metrics leave room for agnosticism for reliability.

TABLE IX
COMPARISON OF OBTAINED RESULT WITH STATE OF THE ART WORKS.

Study	Feature Selection	No. of Feature considered	Classif.	Acc.%
Dwivedi, A.K [7]	No selection	7129	ANN	98
Andini, A et al [9]	PCA (feature Extraction)	7129	PCA with KNN	82
Mallik, P.K et al [8]	No selection	7129	DNN	98.21
Jakelyne Machado Lima Silva, et al. [28]	No selection	7129	Ensemble Method	94.1
Proposed Method	ANOVA	2853	ANN	100

V. CONCLUSION

The examination of gene expression is a useful tool for the rapid and accurate detection of deadly cancers. Numerous studies have been conducted over the years. In order to obtain accurate information, all of the suggested work tries to quickly enhance several accuracy measures. The main goal of our work was to carry out a comparative analysis of three different types of feature selection approaches. The results of accuracy

measures in feature selection processes were encouraging. In this study, another goal was to extract extremely lethal gene expressions. Through the use of feature selection approaches, this was accomplished. To verify the accuracy of various methodologies, six machine learning and, one artificial neural network algorithm were run. Future research can be conducted to determine which genes are most important in causing deadly cancers. This work can be improved further to produce the most precise leukemia classification.

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